

Soil water dynamics around a tree on a hillslope with or without rainwater supplied by stemflow

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[1] A tree can partition rainfall into throughfall and stemflow (SF), causing water to be funneled around the tree base, and can preferentially divert rainwater in soil layers, causing water to be funneled around tree roots. To determine the effects of each on soil water dynamics, we compared soil water dynamics around a tree on a hillslope on the basis of 2 years of field observations before (SF period) and after (non-SF period) intercepting the stemflow of the tree. Additionally, two sprinkling experiments were conducted using different dye tracers to separately identify infiltration pathways derived from throughfall and stemflow. The observation results in the SF period showed irregular variations in soil water content, high soil water storage, and significant saturated zone development in the downslope region from the tree, which were attributed to stemflow concentrated on the downslope side of the tree. Although dramatic variations in soil water dynamics disappeared in the non-SF period, asymmetrical soil water response patterns were also observed, which were mainly attributed to root-induced bypass flow. Focusing on the downslope region in the SF and non-SF periods, the frequency of saturated zone generation at the soil-bedrock interface decreased from 58% to 16%, but the frequency of bypass flow occurrence varied little. Saturated zone generation at the soil-bedrock interface underneath the tree in both the SF and non-SF periods suggests that trees are key locations for rainfall infiltration and that tree-induced saturated zone generation should be considered carefully, even in conditions without stemflow supply.

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1. Introduction

[2] In addition to meteorological factors (e.g., rainfall amount and intensity) and geological factors (e.g., landform, soil depth, and soil texture), vegetation factors (e.g., species and density) have a great influence on spatial variation in soil water dynamics because of heterogeneity in rainfall redistribution and evapotranspiration processes [Breshears *et al.*, 1997; Tromp-van Meerveld and McDonnell, 2006a; Liang *et al.*, 2007]. Precipitation in forests is first intercepted by canopies and is then partitioned into throughfall and stemflow components as diffuse and point inputs, respectively. Thus, the rainwater reaching the forest floor is considerably uneven. Although spatial variation in throughfall contributes to spatial variation in soil water, throughfall patterns may not necessarily be reflected in the soil moisture patterns [Pressland, 1976; Raat *et al.*, 2002]. Excluding spatial variability of topographic or soil physical properties, the different spatial patterns between throughfall and soil moisture would be the influence of localized concentrations of stemflow input [Voigt, 1960; Levina and Frost, 2003; Keim *et al.*, 2006]. At the vegetation community scale, Ludwig *et al.* [2005] reported that

considerably more soil water was stored in small groves of mulga trees than in intergroves where no trees existed, causing a deeper wetting front under the mulga grove than the intergroves. At the single-tree scale, Glover *et al.* [1962] also found a deeper wetting front underneath a desert date tree. Furthermore, Durocher [1990] recorded very rapid water movement and generation of saturated zones underneath each of two sweet chestnut trees and attributed these features to small-scale water inputs by stemflow. Aboal *et al.* [1999] quantified the stemflow of 30 sampled trees belonging to six different species in a laurel forest and found that precipitation could be concentrated up to 12.8 times in the infiltration areas of the trees by stemflow, even though the annual stemflow only represented 6.85% of the gross precipitation. Clearly, the effect of stemflow serving as point inputs of rainwater on soil water dynamics cannot be ignored in forests.

[3] In addition to stemflow, preferential flow also increases the heterogeneity of spatial soil water dynamics. Beven and Germann [1982] indicated that macropores in soil layers may be associated with either living or decayed tree roots and that the structure of macropore systems derived from roots may be very effective at channeling water through the soil layers. On the basis of sprinkling experiments, root observations, and numerical simulations, Lange *et al.* [2009] provided evidence that roots represent the pore system and that root density is closely related to the occurrence of preferential infiltration. Hiramatsu and Kumazawa [2002] reported that the equivalent hydraulic conductivity in areas surrounding tree roots was 1.1–3.2 times the soil matrix conductivity and

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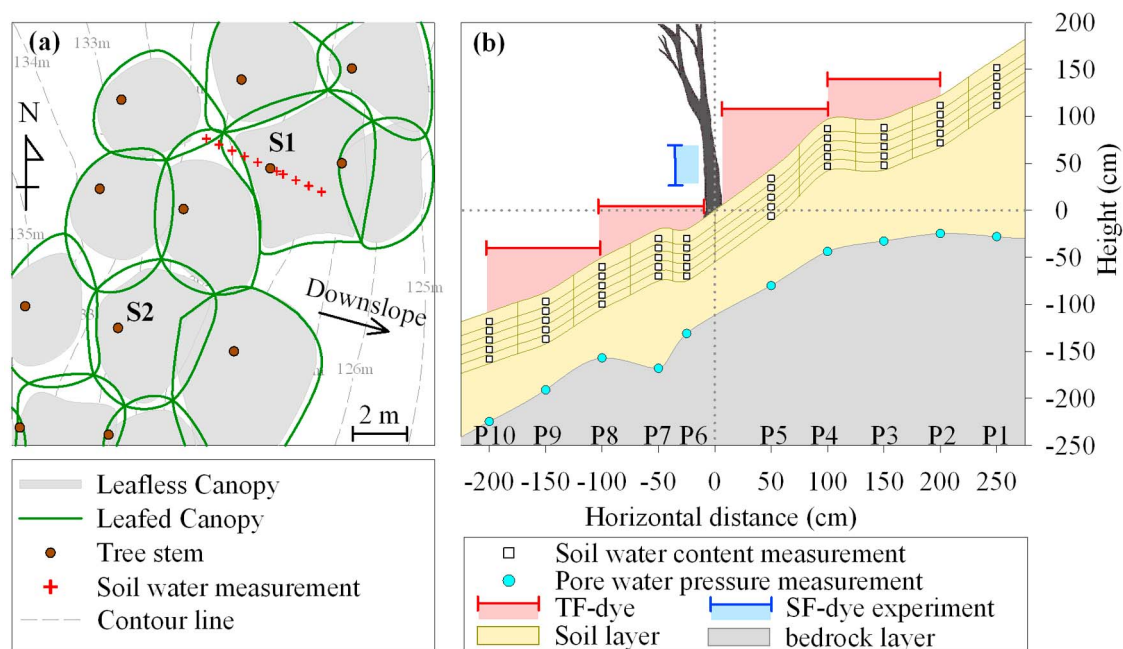


Figure 1. (a) Topography of the observation area, showing the locations of tree stems (*Stewartia monadelphica*), leafed and leafless projected canopy areas, and soil water measurement points. (b) Longitudinal section along the observation line for tree S1, with soil water content and pore water pressure measurement points P1–P10, and the four areas of the throughfall (TF) dye experiment and one area of the stemflow (SF) dye experiment.

that rainwater can be expected to concentrate around roots. Such complex networks of interconnected preferential flow pathways affecting stormflow response at the catchment scale have been revealed by both field observations and conceptual models [Noguchi *et al.*, 1999; Sidle *et al.*, 2000; Lin, 2006]. Thus, the preferential flow pathways created by roots appear to have important implications for soil water dynamics in forested hydrological processes.

[4] On the basis of the above findings, a tree can partition rainfall into throughfall and stemflow, causing water to be funneled around a tree base, and can preferentially divert rainwater in soil layers, causing water to be funneled around tree roots. Johnson and Lehmann [2006] referred to these effects as the double-funneling effect of a tree. For such double funneling, Martinez-Meza and Whitford [1996] demonstrated that root channels are preferential pathways for the movement of stemflow water into soil. Kobayashi *et al.* [2000] measured matric potential at several depths and points distanced from a tree stem and found that concentrated stemflow infiltrated through macropores as bypass flow. However, although previous studies have reported on the double-funneling effect on soil water dynamics, we still lack an understanding of the separate involvement of stemflow and root-induced bypass flow in rainwater infiltration processes.

[5] The purpose of this study was to quantify the effects of stemflow and root-induced bypass flow on soil water dynamics around a tree on a hillslope. We used a 2 year data set of rainfall, stemflow, soil water content, and pore water pressure at the soil-bedrock interface to compare soil water dynamics before and after intercepting stemflow. Occurrences of bypass flow and local generation of a saturated zone

around a tree were analyzed for the two periods. Additionally, field dye experiments were conducted to separately identify infiltration pathways from throughfall and stemflow.

2. Methods and Materials

2.1. Study Area

[6] Observations were conducted on a hillslope (Figure 1a) at the Kamigamo Experimental Station of Kyoto University, located in southern Kyoto Prefecture, central Japan (35°04'N, 135°46'E). The climate of this area is warm temperate. The mean annual air temperature for 1976–2005 was 14.7°C, with highest and lowest monthly averages of 27.4°C (August) and 2.4°C (January), respectively. Mean annual precipitation was 1523 mm, distributed year round, with a peak in summer and only a few centimeters of snow for a short time in winter.

[7] The hillslope had a mean gradient of 28°, with brown forest soil classified as Cambisol underlain by sandstone and slate. There was a small and near-drought channel located 15 m southeast from the hillslope. Tall stewartia (*Stewartia monadelphica*), planted in 1956 (Figure 1a), was the predominant land cover. The genus *Stewartia* includes 21 species that are distributed throughout Japan, Korea, China, and eastern North America [Li *et al.*, 2002]. Of these, tall stewartia is widespread in natural forests in western and southern parts of Japan and is classified by the U.S. Department of Agriculture into hardiness zones 6B through 8B. Like most species of *Stewartia*, tall stewartia is a deciduous broad-leaved tree with upward tilting branches and smooth-exfoliated bark. Tall stewartia exhibited leaf fall in November and regrowth in April in the study area.

2.2. Measurements of Soil Water Dynamics and Stemflow

[8] Measurements of soil water dynamics were conducted for over 2 years, from 20 April 2007 through 31 May 2009. To monitor the soil water dynamics around a tree, we selected a tall *Stewartia* (S1 in Figure 1a; height, 17.47 m; diameter at breast height, 22.3 cm; leafless projected canopy area, 17.39 m²) and delineated a longitudinal observation line from upslope to downslope regions of the tree (Figure 1b). No understory vegetation existed along the observation line, so only the effects of tree S1 on soil water dynamics would be identified. Additionally, no stemflow was observed in the small channel during the observation period, so soil water dynamics would not be influenced by the saturation from the channel. We installed capacitance meters (Sentek, EasyAG-5p) at the following ten points: 250 cm (P1), 200 cm (P2), 150 cm (P3), 100 cm (P4), and 50 cm (P5) upslope from the tree stem and 25 cm (P6), 50 cm (P7), 100 cm (P8), 150 cm (P9), and 200 cm (P10) downslope from the tree stem (Figure 1b). Each capacitance meter consisted of five sensors at depths of 10, 20, 30, 40, and 50 cm; therefore, soil water content at a total of 50 locations was measured. Additionally, we installed tensiometers at the soil-bedrock interface at the same ten points for pore water pressure measurements (Figure 1b). The soil depth to bedrock at each point was determined by penetration tests using a knocking-type cone penetrometer with a 60° bit, a cone diameter of 20 mm, a weight of 2 kg, and a fall distance of 50 cm. From the results of the penetration test, we computed the penetration resistance value as the number of blows required for a 10-cm penetration. Previous studies proposed that a penetration resistance value of 100 may indicate the boundary between soil and bedrock [Okimura and Tanaka, 1980; Yoshimatsu *et al.*, 2002]. The soil depths of all points were estimated to be between 104 and 190 cm (Figure 1b). Pore water pressure measurements were interrupted in winter to avoid water freezing in the tensiometers.

[9] For stemflow measurements, we selected another tall *Stewartia* (S2 in Figure 1a; height, 13.83 m; diameter at breast height, 21.7 cm; leafless projected canopy area, 12.84 m²), located at a similar point on the slope and having a similar tree shape as tree S1. To separately collect the stemflow along the upslope and downslope sides of the trunk of tree S2, we used two tubes cut longitudinally and wrapped spirally around the upslope and downslope sides of the trunk. The flow rates of stemflow (SF) up and SF down were measured using tipping-bucket gauges that tipped at 4 mL (Davis, 7852M) and 500 mL (Ikeda, TQX-500), respectively. From 9 April 2008 through 1 May 2009, we measured SF up and SF down of tree S1 in the same way as tree S2 and installed tipping-bucket gauges at a location 4 m laterally from P10. Thus, rainwater input attributable to stemflow for tree S1 was intercepted from the observation line of soil water dynamics.

[10] The measurements above were simultaneously and automatically recorded at 5 min intervals by a data logger (Campbell, CR1000). Gross rainfall data provided by the Kamigamo Experimental Station of Kyoto University was measured using two tipping-bucket rain gauges (Ikeda RH-5, 0.5 mm per tip) recorded at 5 and 10 min intervals at an open site 112 m from the observation slope.

2.3. Dye Experiments and Soil Sampling

[11] The spatial distribution of soil water in a forest is considerably uneven because of the spatial heterogeneity of

throughfall and point input of stemflow. To remove the spatial heterogeneity of throughfall and to identify infiltration pathways derived from throughfall and stemflow, we conducted two sprinkling experiments with different dye tracers on 12 March 2010.

[12] We used New Coccine (food red number 102) as a dye tracer to identify throughfall (TF) infiltration pathways (TF dye experiment). The dye was mixed into water at a concentration of 5 g L⁻¹, and the red-colored dye solution was sprinkled onto four dye areas containing P2 through P10 using four spray nozzles (Figure 1b). Eighty liters of the dye solution were evenly sprinkled onto each dye area of 40 cm width (lateral direction) × 100 cm length (slope direction) over 32 min, corresponding to throughfall amounts of 50 mm and average intensity of 93.8 mm h⁻¹. At 28 min after the TF dye experiment, we used Brilliant Blue FCF (food blue number 1) as another dye tracer to identify stemflow infiltration pathways (SF dye experiment). The dye was mixed with water at a concentration of 2 g L⁻¹. Using a spray nozzle, 10 L of the bright greenish-blue colored dye solution was sprinkled toward the downslope side of tree stem S1 over 16 min (Figure 1b), corresponding to stemflow intensity of 37.5 L h⁻¹. Soil water contents at all points in both the TF and SF dye experiments were simultaneously recorded at 1 min intervals.

[13] To observe infiltration pathways within the soil profile, we excavated a trench from P1 through P10 in the slope direction and took photographs of the soil profile 2 days after the dye experiments. Then we sampled soil close to each measurement location for soil water content (Figure 1b) using thin-walled steel samplers with a volume of 100 cm³. Fifty undisturbed samples were used for laboratory measurements of soil porosity based on the difference between saturated and absolute dry weights.

2.4. Data Analysis

[14] To determine the effect of stemflow on soil water dynamics, we compared the differences in soil water dynamics between the observation period before (SF period, 20 April 2007 through 8 April 2008) and after (non-SF period, 9 April 2008 through 1 May 2009) intercepting stemflow for tree S1. Both the SF and non-SF periods were approximately 1 year. Each rainfall event in these two periods was analyzed with an individual rainfall event defined as having a rainfall amount more than 0.5 mm, separated from other events by 6 consecutive hours without rain. In regard to meteorological conditions, rainfall amount and rainfall intensity were analyzed using a histogram to compare rainfall characteristics in the SF and non-SF periods. Stemflow amount and intensity of tree S2 in the non-SF period were also analyzed using a histogram. The stemflow measured at the upslope (SF-up) and downslope (SF-down) sides of tree S2 were compared to measurements for tree S1.

[15] To verify the detailed soil water dynamics in the SF and non-SF periods, we compared the spatiotemporal variations in soil water content (θ) and pore water pressure (ψ) between two storm events with the same scale of rainfall amounts in the SF and non-SF periods. The measurements from each of 50 capacitance meter sensors were assumed to represent θ for each element of the mesh system shown in Figure 1b. We calculated water content change ($\Delta\theta$) at all points and depths, defined as the difference between the

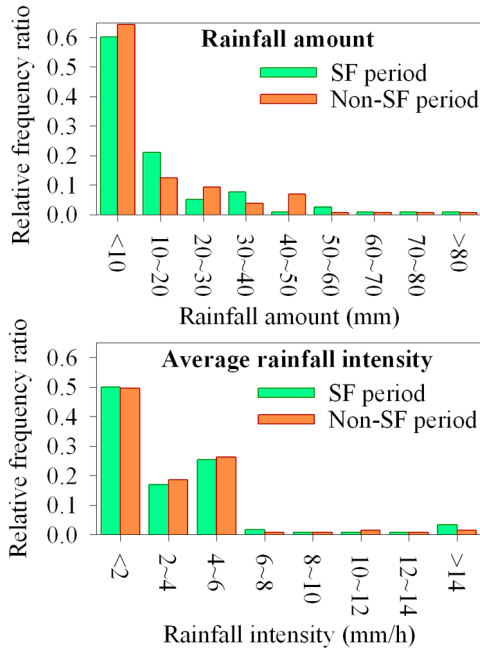


Figure 2. Relative frequency histograms of open-area rainfall amount and average rainfall intensity for rainfall events in the SF and non-SF periods.

current water content and the initial water content at the start of a rainfall event. We also calculated the value of the hydraulic head (ϕ) at the soil-bedrock interface as the sum of the ψ value and the height of the soil-bedrock interface (i.e., elevation head) shown in Figure 1b.

[16] To analyze the detailed variation in θ around the tree, we first evaluated the absolute accumulated variation in soil water content ($\Sigma\Delta\theta_m^j$) at all measurement locations m ($m = 1, 2, \dots, 50$) for the j th rainfall event that caused an obvious θ increase of more than $0.005 \text{ cm}^3 \text{ cm}^{-3}$ in at least one location by

$$\Sigma\Delta\theta_m^j = \Sigma|\theta_m^{j,t+1} - \theta_m^{j,t}|, \quad (1)$$

where $t + 1$ and t are the time steps at 5 min measurement intervals in the j th rainfall event. Then we used standard score (Z score) to remove the factor of rainfall magnitude so that $\Sigma\Delta\theta_m^j$ was rescaled by

$$\Sigma\Delta\theta_m^{j*} = (\Sigma\Delta\theta_m^j - \mu^j)/\sigma^j, \quad (2)$$

where μ^j and σ^j are the mean and standard deviation of $\Sigma\Delta\theta_m^j$ ($m = 1, 2, \dots, 50$) for the j th rainfall event, respectively. First, the spatial distribution of the rescaled accumulated variation, $\Sigma\Delta\theta^*$, at each measurement location calculated in the SF period was compared to the non-SF period. Second, we evaluated the occurrence of bypass flow at each point on the basis of the vertical response change of θ for each rainfall event. The occurrence of bypass flow is recognized when soil water responds to rainwater inputs earlier in deeper soil layers than in shallower layers [Kobayashi *et al.*, 2000]. Thus, we investigated the response change from depths of 10–50 cm at each point for each rainfall event in the SF and non-SF periods and presumed the occurrence of bypass

flow when an earlier water content response occurred in a deeper layer. Third, we quantified the maximum increase in soil water storage ($\Delta S_{\max}$) at each point for the rainfall events that caused θ increase of more than $0.005 \text{ cm}^3 \text{ cm}^{-3}$. For each point, the $\Delta S_{\max}$ from the surface through to a depth of 55 cm was calculated by $\Delta\theta$ measured at depths of 10, 20, 30, 40, and 50 cm. Finally, we investigated saturated zone generation on the basis of the maximum value of ψ measured at each point for each rainfall event. Here we assumed that an obvious ψ increase of more than 5 cm indicated a response to the rainwater input.

3. Results

3.1. General Trends in Rainfall and Stemflow Characteristics

[17] There were 118 and 129 rainfall events observed in the SF and non-SF periods, respectively. The relative frequency of rainfall amount and average rainfall intensity was quite similar in both the SF and non-SF periods (Figure 2). In the SF and non-SF periods, event rainfall amounts ranged from 0.5 to 93 mm and from 0.5 to 113 mm, respectively, and average rainfall intensity ranged from 0.2 to 31.7 mm h^{-1} and 0.2 to 48 mm h^{-1} , respectively. Thus, little difference was observed in the precipitation conditions between the SF and non-SF periods.

[18] The relative frequency of stemflow amount and average intensity of tree S2 was similar (Figures 3a and 3b), ranging from 0 to 426.5 L and from 0 to 135.1 L h^{-1} , respectively. For both tree S1 and tree S2, SF down was considerably greater than SF up (Figure 3c). The total volume of SF down was 43 and 65 times greater than that of SF up for trees S1 and S2, respectively. The total stemflow (the sum of SF down and SF up) of S1 was slightly higher (1.08 times) than that of S2 (Figure 3d), suggesting almost the same stemflow generation rate for trees S1 and S2. Other detailed information of stemflow (e.g., seasonal variations) was described in our previous study [Liang *et al.*, 2009b].

3.2. General Trends in Soil Water Dynamic Changes

[19] Figure 4 shows the 2 year time series of rainfall, θ , and ψ including the SF period, non-SF period, and 1 month after the non-SF period when stemflow was again input into soil layers beginning in May 2009. In both the SF and non-SF periods, rainfall amount and rainfall intensity increased in summer and decreased in winter because of intensive frontal rainfalls and convective storms that occurred in summer.

[20] In the SF period, θ values obviously increased at all depths in the upslope region when rainfall intensity was high, with responses at P3 larger than those at other points. When rainfall intensity was low (e.g., October–December 2007), responses of θ were confined to the upper layers of 10–20 cm depths. In periods of no precipitation, θ decreased greatly in summer but only gradually in winter because of active evapotranspiration in summer. In the downslope region, P8–P10 showed similar response patterns of θ to the points in the upslope region, but P6 and P7 showed dramatic changes in θ even for low rainfall intensity events. Values of ψ increased or decreased slightly at the points in the upslope region but decreased markedly in the downslope region. Compared to the upslope points, the points in the downslope region showed sharp responses of ψ in all of the SF period,

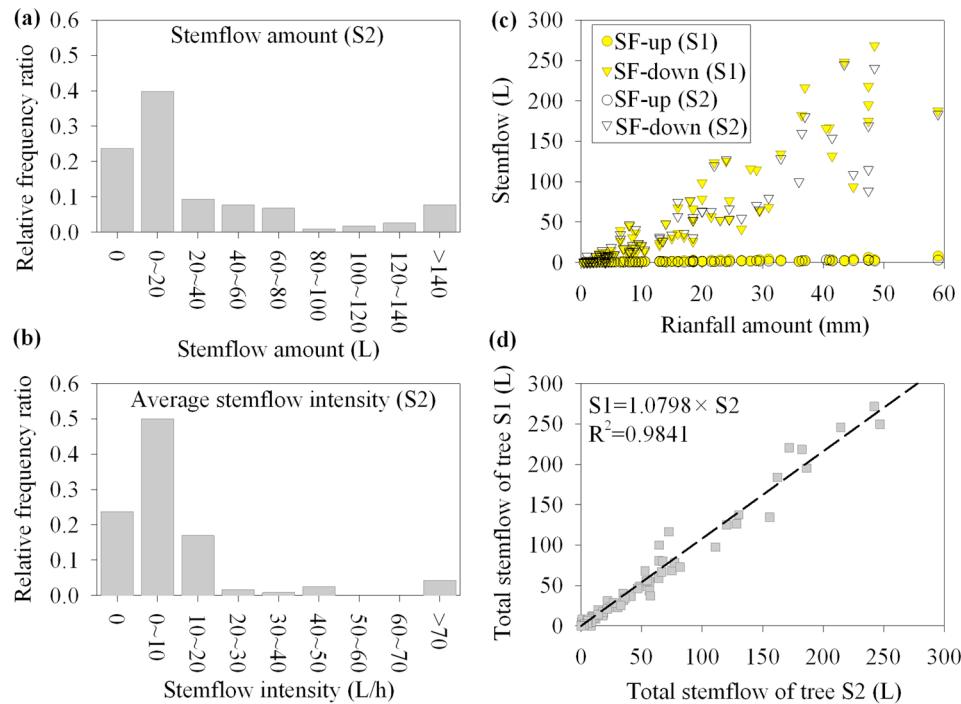


Figure 3. Relative frequency histograms of (a) total stemflow amount and (b) average stemflow intensity of tree S2 and (c) relationships between open-area rainfall and stemflow measured along the upslope (SF-up) and downslope (SF-down) sides and (d) total stemflow of tree stems S1 and S2 for each event in the non-SF period.

even for P8–P10, which had a similar response pattern as θ to points in the upslope region. In particular, positive values of ψ were often measured at P6 and P7, indicating transient saturated zones at the soil-bedrock interface. However, few positive values of ψ were measured at points in the upslope region.

[21] In the non-SF period, response patterns of θ seemed similar at all points in both upslope and downslope regions and varied with rainfall intensity. However, response patterns of ψ remained different between upslope and downslope points. Responses of ψ at downslope points were still sharper than those of upslope points, with P6 showing positive ψ values for several events. After the non-SF period in which stemflow was again input into soil layers, spiked variations in θ at some depths and frequent positive values of ψ were measured at P6 and P7, like in the SF period.

3.3. Soil Water Dynamic Changes in Two Storm Events

[22] Figure 5 shows temporal and spatial variations in soil water dynamics around tree S1 for a storm event on 25 May 2007 in the SF period. For the event, accumulated rainfall and stemflow of tree S2 were 62 mm and 280.8 L, respectively, and maximum rainfall and stemflow intensities were 2.5 mm per 10 min and 6.6 L per 5 min, respectively. Stemflow occurred rapidly 20 min after rainfall began and showed a similar response to rainfall (Figure 5a). On the basis of the results in Figure 3d, we assumed that stemflow measured at tree S2 could represent the stemflow rate at tree S1.

[23] At P3 (upslope of tree S1, Figure 5a), increases in θ were only observed at depths of 10–30 cm and the response

of ψ was not measured, indicating that infiltrated rainwater was confined within the upper soil layers. At P6 and P7 in the downslope region, large increases in θ were observed at each depth, particularly a spiked response at 50 cm depth at P6 and 30 cm depth at P7. The value of ψ rapidly and largely increased at P6, where a positive value greater than 20 cm was observed during a large part of the event. P7 also showed a rapid and large increase in ψ , although its response started 200 min later than P6. For spatial soil water dynamics at 50 min (Figure 5b), the accumulated rainfall was 3 mm, with no obvious changes in $\Delta\theta$ and ϕ observed at any point except for P6. At 565 min, rainfall and stemflow intensities reached the maximum for this storm event. At P6 and P7, $\Delta\theta$ increased greatly, especially at depths of 50 cm at P6 and 30 cm at P7; however, the increases of $\Delta\theta$ at other points were limited to the upper soil layers of 10–20 cm. In the downslope region, ϕ from P6–P9 was much larger than for the soil-bedrock interface; however, there were no obvious changes in ϕ in the upslope region except for P5. At the end of the event (890 min), obvious increases in $\Delta\theta$ were measured above a depth of 40 cm at all points, but increases in ϕ were limited to the points in the downslope region and P5. The rise of ϕ at P5 could be attributed to expansion of a saturated zone from the downslope to upslope regions rather than vertical expansion of the wetting front.

[24] Figure 6 shows temporal and spatial variations in soil water dynamics for a storm event on 17 April 2008 in the non-SF period when stemflow of tree S1 was intercepted. During the event, accumulated rainfall and stemflow of trees S1 and S2 were 65 mm, and 245.2 L and 214.5 L, respectively, and maximum rainfall and stemflow intensities of

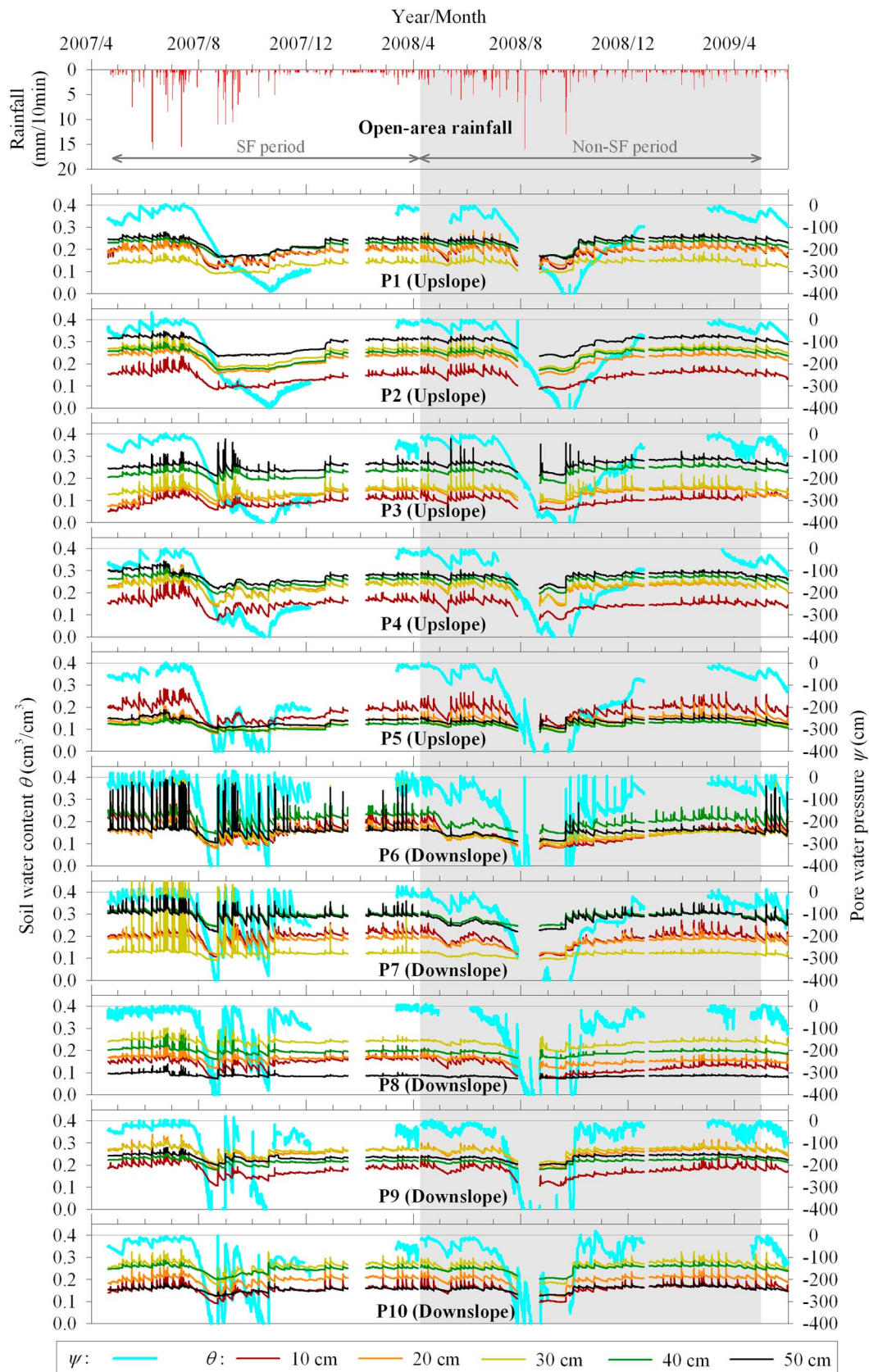


Figure 4. Open-area rainfall, soil water content (θ), and pore water pressure (ψ) at 10 min intervals at the 10 measurement points P1–P10 in regions upslope and downslope of tree S1. Interruptions within data lines indicate missing data periods. The shaded period indicates the non-SF period in which stemflow from tree S1 was intercepted.

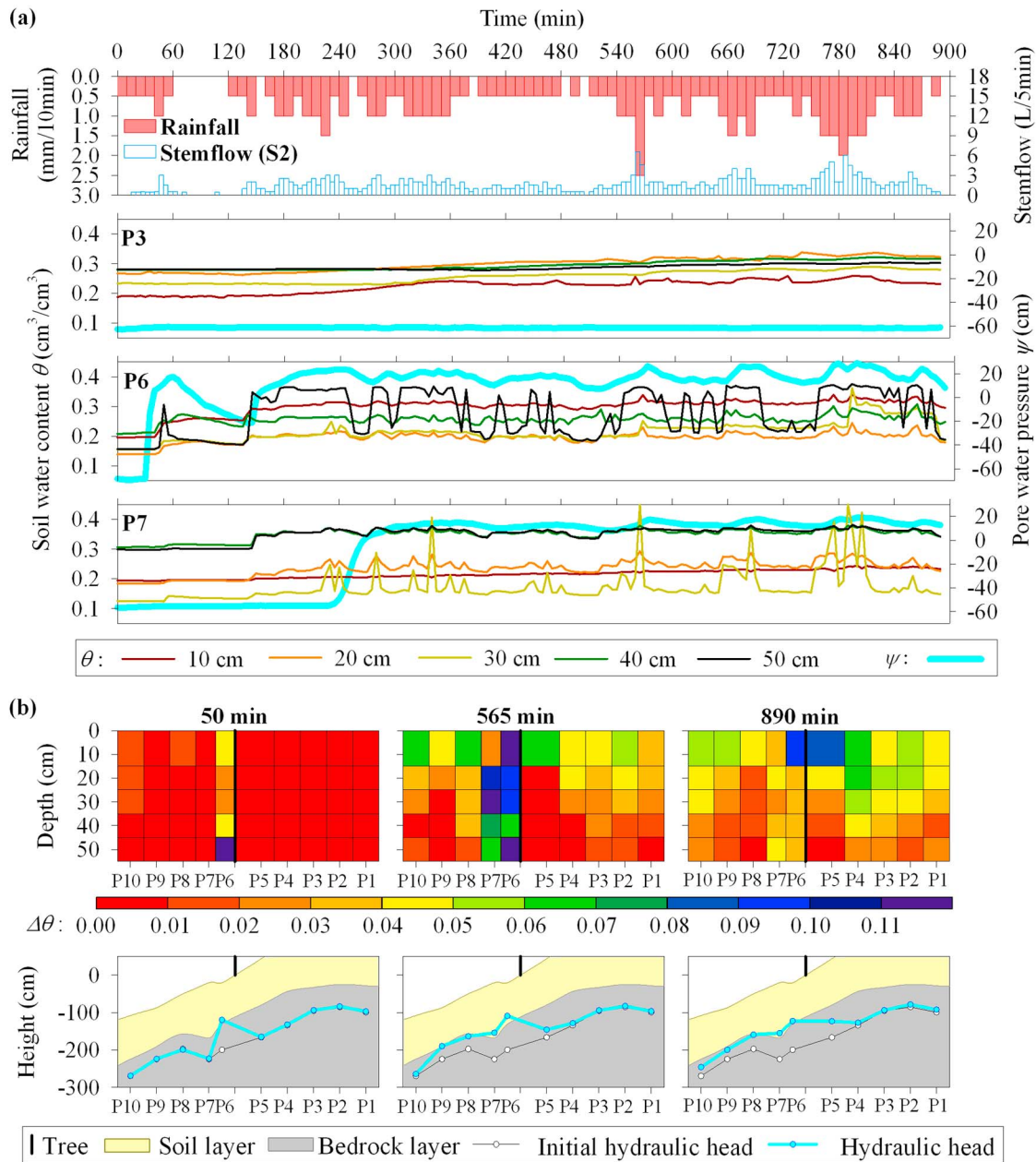


Figure 5. (a) Temporal variations in open-area rainfall, stemflow of tree S2, soil water content (θ), and pore water pressure (ψ) at P3, P6, and P7 for a storm event on 25 May 2007 before intercepting stemflow from tree S1. (b) Spatial variations in soil water content change ($\Delta\theta$) and hydraulic head (ϕ) distribution at 50, 565, and 890 min for the event. In Figure 5a, all data are shown at 5 min intervals except for rainfall, which is shown at 10 min intervals. $\Delta\theta$ is defined as the difference between the current water content and the initial water content observed at the start of the storm event, ϕ is computed as the sum of the observed ψ and the height of the soil-bedrock interface, and initial ϕ is the value observed at the start of the storm event.

trees S1 and S2 were 2.5 mm and 8.6 L and 7.6 L per 5 min, respectively. Soil water dynamics, θ and ψ , gradually increased or decreased with rainfall intensity and showed a very similar response pattern at P3, P6, and P7 (Figure 6a). At 240 min when the accumulated rainfall and intercepted stemflow were 10 mm and 23.6 L (Figure 6b), respectively, increases in $\Delta\theta$ were limited to a depth of 10 cm, except for at 40-cm depth at P6 where a slight $\Delta\theta$ of 0.02 was found; meanwhile, no response in ϕ was observed at any point. At

1320 min, rainfall intensity and intercepted stemflow reached the maximum for this storm event. Obvious increases in $\Delta\theta$ were observed at depths of 10–50 cm at P3 and 10–30 cm at other points, while minimal rises of ϕ to the surface were observed at all points. At 1410 min when rainfall was interrupted, $\Delta\theta$ decreased when compared to conditions at 1320 min but ϕ rose continuously and reached the maximum at each point, indicating a response time lag of rainwater reaching the soil-bedrock interface. P7 showed the maximum

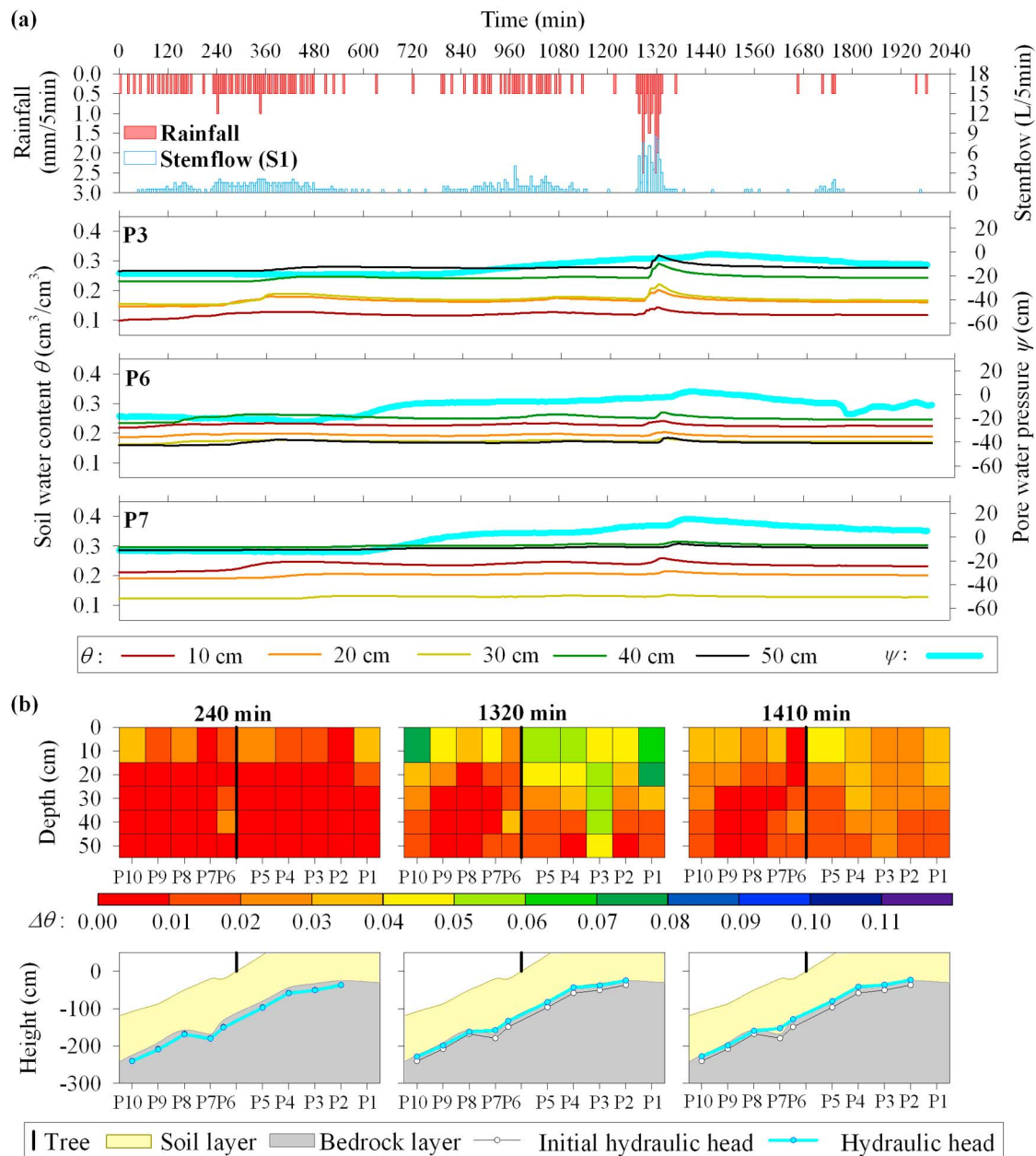


Figure 6. (a) Temporal variations in open-area rainfall, intercepted stemflow of tree S1, soil water content (θ), and pore water pressure (ψ) at P3, P6, and P7 for a storm event on 17 April 2008 after intercepting stemflow of tree S1. (b) Spatial variations in soil water content change ($\Delta\theta$) and hydraulic head (ϕ) distribution at 240, 1320, and 1410 min for the event. In Figure 6a, all data are shown at 5 min intervals. In Figure 6b, no data are shown for P1 because of instrument failure. Other details are as described for Figure 5.

ψ of 15.3 cm for all points in this event, which was likely attributable to a local depression in the bedrock topography.

[25] Comparing the soil water dynamics in these two storm events with the same scale of rainfall amount, spiked response waves were not found in an event without stemflow, although the maximum rainfall intensity in the non-SF storm event was greater than the SF storm event. This finding indicated that spatial and temporal variations in soil water content and saturated zone generation were greatly affected by the stemflow that concentrated on the downslope side of tree trunks (Figure 3c).

3.4. Infiltration Pathways Identified by Dye Experiments

[26] Comparing rainfall and stemflow characteristics observed in this study (Figures 2, 3a, and 3b), the sprinkling intensities of both the TF and SF dye experiments were very high, representing infiltration processes for a storm event. During TF dye experiments, the response was gradual and increased from surface layers through deeper layers at all sprinkled points except for P3 and P6 (Figure 7), where rapid and large increases of θ were measured at deeper depths.

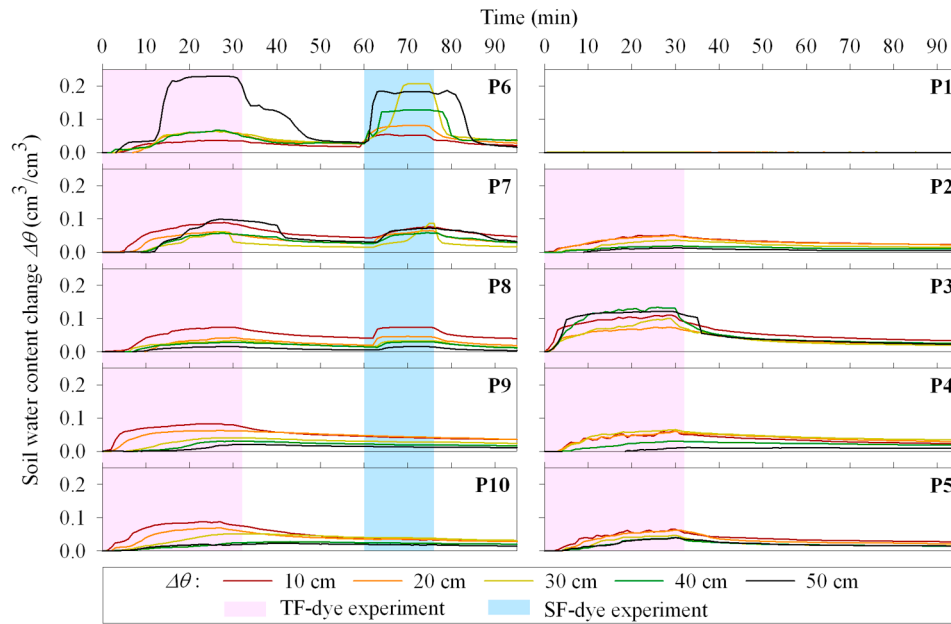


Figure 7. Soil water content change ($\Delta\theta$) for the throughfall (TF) and stemflow (SF) dye experiments. The TF dye experiment was conducted in the region downslope from P1 and the SF dye experiment was conducted at the downslope side of the tree trunk (Figure 1b).

During the SF dye experiment, obvious increases of θ were measured at P6–P8. In particular, large increases of θ were measured at 30–50 cm depths at P6. On the basis of the response from depths of 10–50 cm, bypass flow likely occurred at P6 in the TF dye experiment and at P6 and P7 in the SF dye experiment.

[27] Figure 8 shows dye-stained areas after the TF dye (red stained areas) and SF dye (greenish-blue stained areas) experiments. For TF dye experiments, soil was well stained to depths of 0–10 cm but was minimally stained in the deeper layers in the downslope and upslope regions (Figures 8a and 8b). Additionally, there were some locally stained areas in deeper layers (e.g., P3) where roots, rocks, or cracks were found (Figure 8e), which likely reflected the occurrence of preferential flow. For SF dye experiments, soil in the surface area of 0–5 cm depths and deeper areas around thick and fine roots at P6–P8 were well stained (Figure 8a). At the point underneath the trunk, stained areas were found around fine roots at 0–60 cm depths (Figure 8d). At P6–P8, concentrated stained areas were found at 30–50 cm depths (Figure 8a), where thick roots with diameters of 10–20 cm existed (Figure 8c), but not at depths above 30 cm except for the surface area (Figure 8a). This result suggests that only a small amount of SF dye water flowed along the soil surface and that a great amount of SF dye water was diverted by vertical and horizontal roots and infiltrated into deeper soil layers (Figure 8f), indicating the occurrence of bypass flow.

[28] The results in Figures 7 and 8 point to different infiltration pathways for throughfall and stemflow. Throughfall infiltrated as matrix flow vertically into soil layers and caused small θ increases at some points (e.g., P2 and P4). Throughfall also infiltrated as preferential flow that caused large θ increases at other points (e.g., P3, P6, and P7). In contrast, stemflow functioned as bypass flow that vertically and horizontally infiltrated into soil layers and caused larger θ increases at deeper depths than at shallower depths (e.g., P6–P8).

3.5. General Trends in Generation of Bypass Flow and Saturated Zones Around a Tree

[29] There were 85 and 72 rainfall events that caused responses in soil water content (more than $0.005 \text{ cm}^3 \text{ cm}^{-3}$) in the SF and non-SF periods, respectively. Figure 9 shows vertical distributions of $\Sigma\Delta\theta^*$ (equation (2)). The distribution of $\Sigma\Delta\theta^*$ indicates where soil water varied greatly among the 50 measurement locations. In the SF period, $\Sigma\Delta\theta^*$ decreased with depth at all points except for P6 and P7, where irregular vertical distributions were observed. P6 and P7 showed a wide range of $\Sigma\Delta\theta^*$, in which almost all depths had positive and higher averages of $\Sigma\Delta\theta^*$ than other points. This finding indicates that increases or decreases of θ at P6 and P7 were greater than the mean value of all 50 measurement locations (i.e., $\Sigma\Delta\theta^* = 0$). In particular, the maximum $\Sigma\Delta\theta^*$ among all 50 measurement locations was found at 50 cm depth at P6, implying where there was an active infiltration pathway in the SF period. In the non-SF period, the range and value of $\Sigma\Delta\theta^*$ roughly decreased with depth at all points except for P6 and P7, where irregular vertical distributions were still found. The 40 cm depth at P6 showed the maximum $\Sigma\Delta\theta^*$ among all 50 measurement locations, indicating that the active infiltration pathway still existed in the non-SF period. Comparing soil porosity and $\Sigma\Delta\theta^*$ (Figure 9), the relationship was not statistically significant at P6 and P7 ($p > 0.05$) but was significant at the other points ($p < 0.05$) in both the SF and non-SF periods. Such a lack of $\Sigma\Delta\theta^*$ dependence on soil porosity or depth at P6 and P7 in both the SF and non-SF periods was likely due to rainwater infiltration through preferential pathways as bypass flow.

[30] Figure 10 shows the frequency of bypass flow occurrence at each point during the SF and non-SF periods. In the SF period, θ responded to 72% and 59% of all rainfall events at P6 and P7, respectively. The response ratio was relatively higher in the downslope region than in the upslope

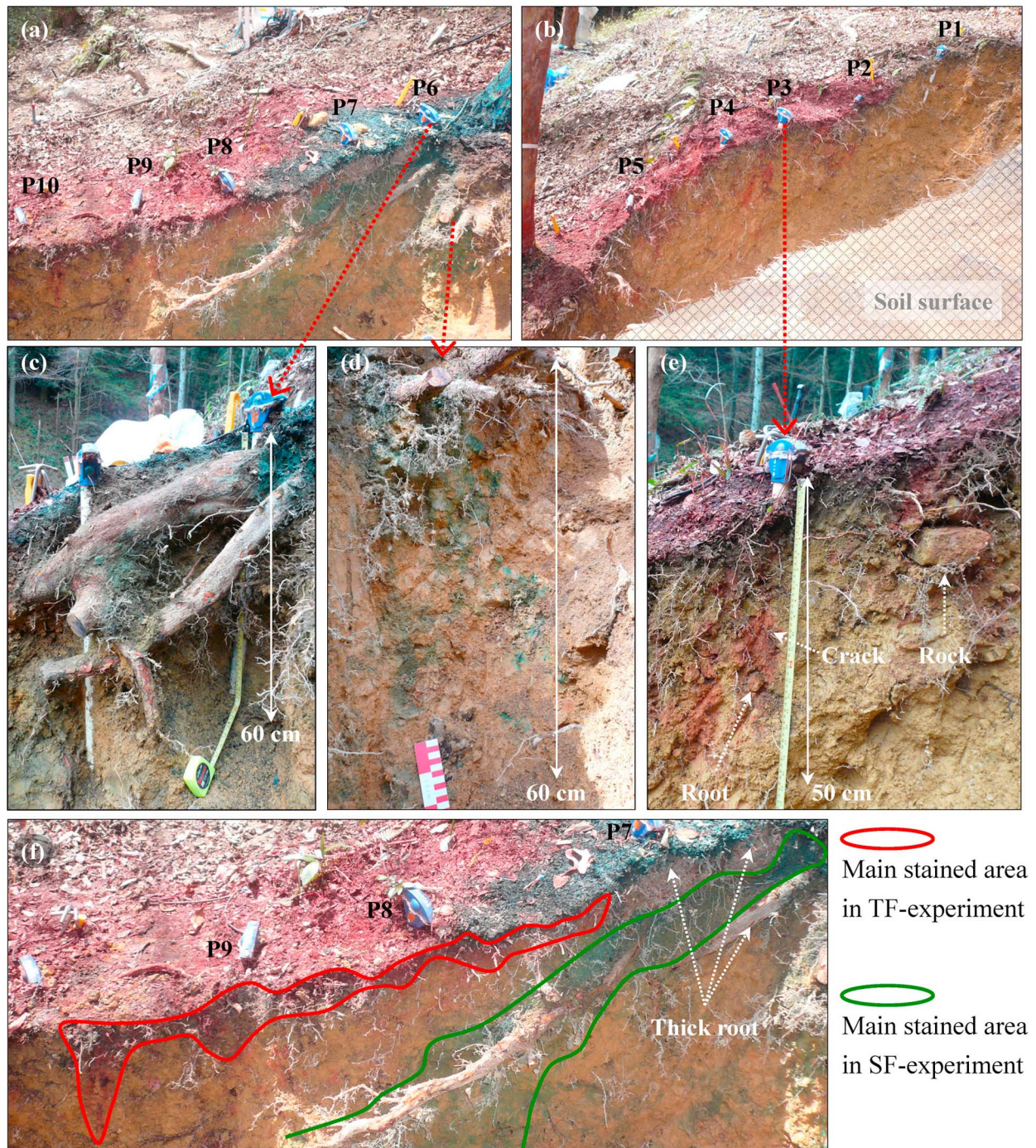


Figure 8. Photographs showing dye-stained areas in the (a) downslope and (b) upslope regions; detailed profiles underneath (c) P6, (d) the tree trunk, and (e) P3; and (f) marked main stained areas in the downslope region. Red and greenish-blue stained areas resulted from the throughfall (TF) and stemflow (SF) dye experiments, respectively. The hatched area in Figure 8b is the surface on the opposite side of the dye-stained profile.

region, which showed average ratios of 57% and 46%, respectively (Table 1). At P7, bypass flow occurred for 65% of the rainfall events that caused a response in soil water content. This ratio was high at other points in the downslope region (27%–48%) as well but was small at points in the upslope region (7%–20%) except for P5 (35%). In the non-SF period, a smaller difference in the response ratio was observed with averages of 45% and 42% in the downslope

and upslope regions, respectively (Table 1). For the rainfall events that caused a response in soil water content, the points in the downslope region showed a high occurrence ratio of bypass flow (44%–59%) except for P9 (4%). The ratio was small at the points in the upslope region (14%–32%). From the SF period to the non-SF period, response ratios clearly decreased at P6–P8 but did not clearly decrease at the other points in the downslope and upslope regions. Although the

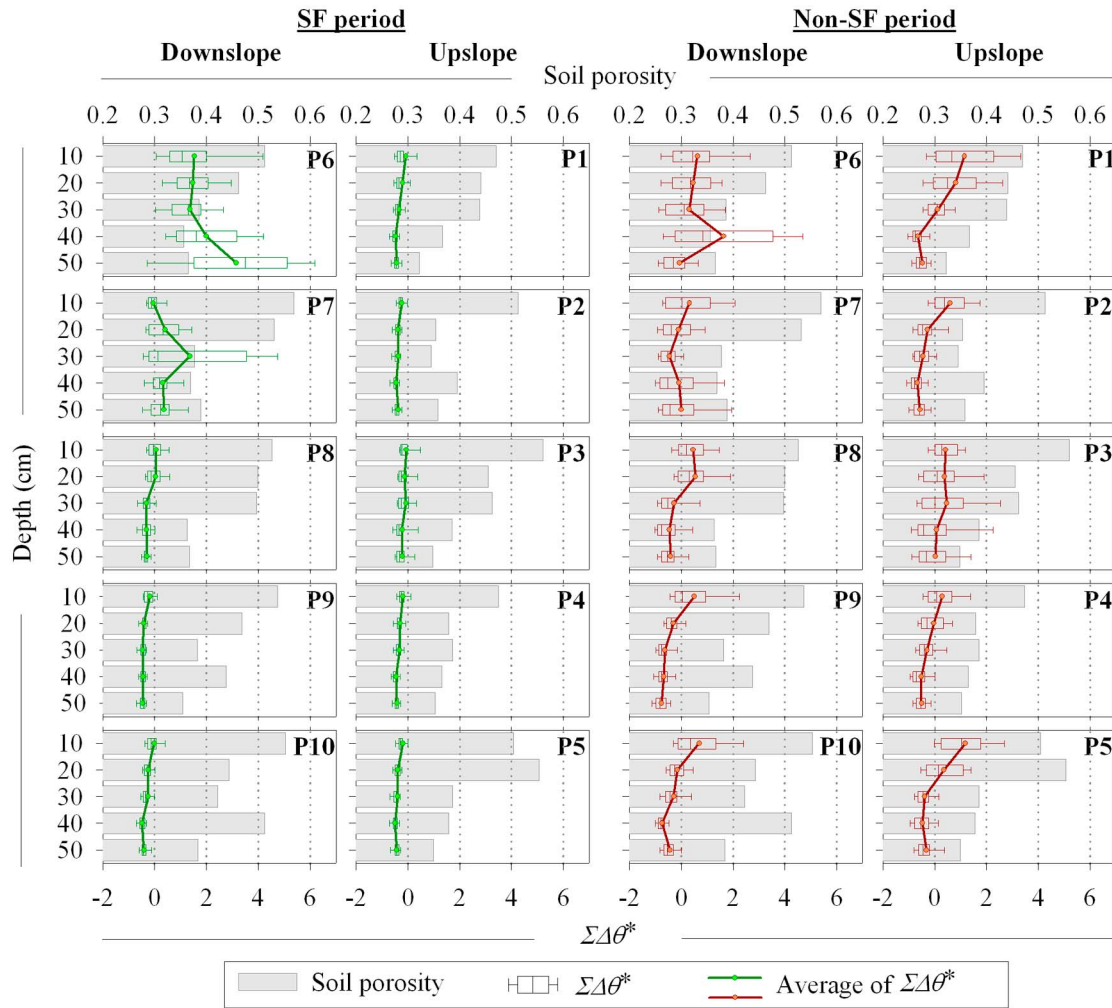


Figure 9. Vertical distributions of soil porosity (bar plot), the variation in water content ($\Sigma\Delta\theta^*$, box plot) as calculated by equation (2), and the average of $\Sigma\Delta\theta^*$ at each point during the SF and non-SF periods. Boundaries of the box indicate the 25th and 75th percentiles, the line within the box indicates the median value, and error bars indicate the 10th and 90th percentiles.

occurrence ratio of bypass flow at all points varied little from the SF to non-SF periods, bypass flow more frequently occurred at points in the downslope region than in the upslope region in both the SF and non-SF periods (Table 1).

[31] Figure 11 presents $\Delta S_{\max}$ and accumulated open-area rainfall from the start of the event to the time when $\Delta S_{\max}$ was recorded for each rainfall event. According to the theory of vertical infiltration, soil water storage should be less than rainwater input. However, $\Delta S_{\max}$ greater than 100% of the accumulated open-area rainfall (SGAR) was found. In the SF period, SGAR was observed for 94% and 65% of all responding rainfall events at P6 and P7, respectively. The ratios were moderately low at P8 (32%), P3 (22%), and P10 (18%) but very low at the other points (0%–7%). In the non-SF period, SGAR was relatively frequently measured at P6 (22%), P7 (17%), and P3 (13%). The ratios were very low at the other points (0%–8%). Although the frequency of SGAR greatly decreased from the SF to non-SF periods in the downslope region, SGAR in the downslope region occurred more frequently than in the upslope region during both the SF and non-SF periods (Table 1).

[32] Figure 12 shows the frequency of the development of a saturated zone at each point during the SF and non-SF periods. In the SF period, ψ at the soil-bedrock interface responded to 62% and 46% of all rainfall events at P6 and P7, respectively. The ratio was also high at P8–P10 (36%–42%) and P5 (34%) but small at P1–P4 in the upslope region (12%–18%). Except for P10, generation of a saturated zone was frequently observed at the points in the downslope region, particularly at P6 and P7, where saturated zone generation was observed for 93% and 88% of the rainfall events that caused a response in ψ , respectively. At the points in the upslope region, the generation of a saturated zone was infrequently observed. In the non-SF period, the response ratio was higher at P6 and P7 (40% and 30%) but smaller at the other points (8%–27%). For the rainfall events that caused a response in ψ , a high frequency of saturated zone development was still recorded at P6 (44%), although ψ more than 20 cm was not observed in this period. The frequency of saturated zone development was very small at the other points (0%–13%). Thus, a saturated zone was generated

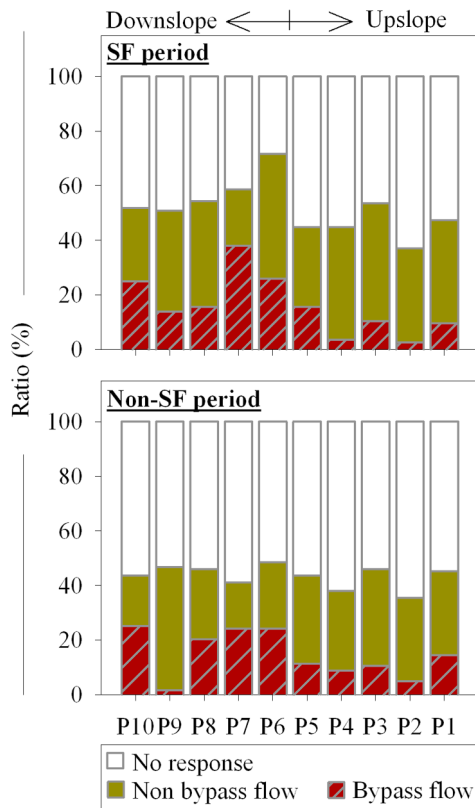


Figure 10. The ratio of rainfall events in which bypass flow occurred to all rainfall events at each point in the SF and non-SF periods. Bypass flow occurrence was determined on the basis of the response change of soil water content from the surface to deeper layers in each rainfall event.

more frequently in the downslope region than in the upslope region in both the SF and non-SF periods (Table 1).

[33] In summary, Figures 10, 11, and 12 show asymmetrical soil water dynamics between regions downslope and upslope from the tree in both the SF and non-SF periods. Focusing on the variation in the downslope region from the SF to non-SF periods, the frequency of SGAR decreased from 43% to 10%, the frequency of saturated zone development decreased from 58% to 16%, but the frequency of bypass flow occurrence changed little from 41% to 42% (Table 1). Although stemflow did not increase the frequency of bypass flow occurrence, it did increase θ responses and activated specific infiltration pathways (e.g., at 50 cm depth at P6 in Figure 9).

4. Discussion

4.1. The Effect of Stemflow on Hillslope Hydrological Processes

[34] Regarding the soil water dynamics in the SF period, the points close to the downslope side of the tree (i.e., P6 and P7) showed larger variations, irregular distributions of $\Sigma\Delta\theta^*$ (Figure 9), and high $\Delta S_{\max}$ greater than accumulated open-area rainfall (Figure 11). These points are thought to be dominated by local and rapid infiltration of bypass flow (Figure 10) activated by stemflow. The other points, how-

ever, had moderate responses, a regular distribution of $\Sigma\Delta\theta^*$ with depth, and a $\Delta S_{\max}$ less than the accumulated open-area rainfall and were thus dominated by a slow expansion of the wetting front. This asymmetrical infiltration pattern between the downslope and upslope regions from a tree caused an asymmetrical saturated zone development at the soil-bedrock interface (Figure 12) that was only frequently observed in the downslope region.

[35] This asymmetrical hydrological process was attributed to an asymmetrical rainwater supply of stemflow that concentrated on the downslope side of the tree trunk (Figure 3c). In our previous study [Liang *et al.*, 2009b], we found that the amount of SF down was 12–132 times greater than SF up on the basis of 2 year measurement results for six tall *Stewartia* and suggested that this difference was little affected by meteorological conditions (i.e., rainfall amount, rainfall intensity, wind speed, and wind direction) but resulted from the uneven area between the upslope and downslope sides of the canopy and from asymmetrical stemflow pathways between the upslope and downslope sides of the trunk due to downslope tilting of the tree trunk. In general, unlike trees growing on flat land, trees growing on a steep hillslope incline toward the slope and are more or less “S” shaped [Schweingruber, 1996, p. 276], causing uneven canopy architecture and downslope tilting of a tree trunk. Thus, this asymmetrical generation of stemflow is probably common for trees growing on a hillslope. Although many studies have observed rapid water movement and generation of a saturated zone underneath the tree or vegetation community [Glover *et al.*, 1962; Durocher, 1990; Ludwig *et al.*, 2005], these types of asymmetrical hydrological processes have not been reported.

[36] In addition to asymmetrical distribution of water supply, preferential infiltration pathways around the tree also increased the heterogeneity of soil water dynamics. Dye tracer and image analyses have often been used to identify infiltration pathways [Kulli *et al.*, 2003; Sander and Gerke, 2007]. Noguchi *et al.* [1999] conducted a dye experiment and illustrated that infiltrated flow impeded by living roots can be diverted around the perimeter of roots, causing generation of subsurface flow and saturated zones in the soil matrix. Martinez-Meza and Whitford [1996] and Li *et al.* [2009] sprinkled a dye tracer on the soil surface around the base of a trunk and demonstrated that root channels of desert shrubs were preferential pathways for the movement of stemflow water into soil. Although the previous studies mentioned

Table 1. Average Frequencies of Soil Water Content Response, Bypass Flow Occurrence, Maximal Soil Water Storage Greater Than Accumulated Open-Area Rainfall (SGAR), Pore Water Pressure Response, and Saturated Zone Generation in the Downslope and Upslope Regions From the Tree in the SF and Non-SF Periods

	SF Period		Non-SF Period	
	Downslope	Upslope	Downslope	Upslope
Soil water content response	57%	46%	45%	42%
Bypass flow occurrence	41%	18%	42%	24%
SGAR	43%	5%	10%	5%
Pore water pressure response	45%	19%	29%	15%
Saturated zone generation	58%	15%	16%	4%

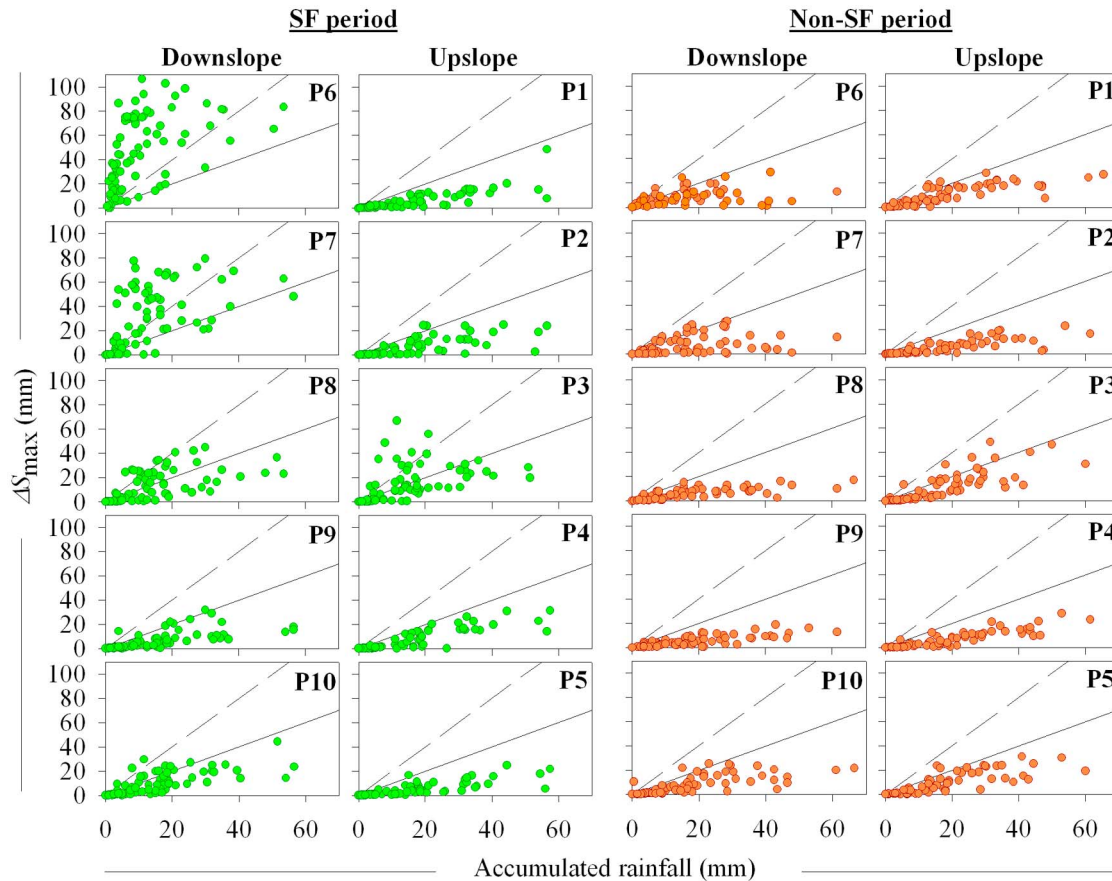


Figure 11. Relationship between the maximum increase in soil water storage from the surface through 55 cm depth ($\Delta S_{\max}$) and accumulated open-area rainfall from the start of the event to the time when $\Delta S_{\max}$ was recorded for the SF and non-SF periods. Dashed and solid lines indicate ratios of 200% and 100%, respectively.

above demonstrated preferential flow pathways derived from roots in soil layers, the application of only one dye tracer could not represent the different infiltration characteristics of throughfall and stemflow in a forested stand. In this study, we conducted sprinkling experiments using two different dye tracers and successfully represented the different infiltration pathways of throughfall and stemflow (Figure 8f). In the TF dye experiment, most of the sprinkled water that vertically infiltrated into soil layers was limited to 0–10 cm depths. Besides shallow infiltration, deeper infiltration attributed to preferential flow was also found at some points. For example, the larger responses (Figure 4) and $\Delta S_{\max}$ (Figure 11) were observed at P3 than at other points in the upslope region. In the SF dye experiment, the sprinkled water significantly bypassed surface layers and horizontally infiltrated into soil layers around the thick root (Figure 8a), showing different infiltration pathways from the TF dye experiment. Because of the overlap of the two dye colors, the stained area was not easily viewable at P6 (Figures 8a and 8c). We also used the supplemental $\Delta\theta$ data for the two sprinkling experiments to verify throughfall and stemflow infiltration processes. We found earlier and larger $\Delta\theta$ increase at a deeper depth at P6 in the TF and SF dye experiments (Figure 7), indicating that bypass flow occurred in both experiments. However, the depths with large increases of θ were not the same for the

TF and SF dye experiments, suggesting that stemflow did activate different pathways than throughfall. Although the results in Figures 7 and 8 only could be representative for a storm event, we presume that different supplied intensity of throughfall and stemflow could activate different pathways because of a wide range of $\Sigma\Delta\theta^*$ observed at P6 and P7 (Figure 9). Thus, use of sprinkling experiments with different dye tracers is an effective method to separately identify infiltration pathways derived from throughfall and stemflow.

[37] Although previous studies have indentified stemflow infiltration pathways by dye experiments [Martinez-Meza and Whitford, 1996; Li et al., 2009], sprinkling a dye tracer on the soil surface around the base of the trunk (i.e., similar to the TF dye experiment in this study) would not completely represent the connectivity from a tree trunk to the root system in stemflow infiltration processes. We suggest that sprinkling water toward a tree trunk applied in this study is a better method to represent stemflow infiltration. In addition, we also suggest that inputting stemflow to the surface infiltration area (i.e., local soil surface area around a tree stem) in the numerical model proposed by Tanaka et al. [1996] would not represent the bypassing characteristic of stemflow, which could represent only local and vertical infiltration characteristics. Considering the bypass characteristic of stemflow infiltration, we proposed a new infiltration model in our previous study

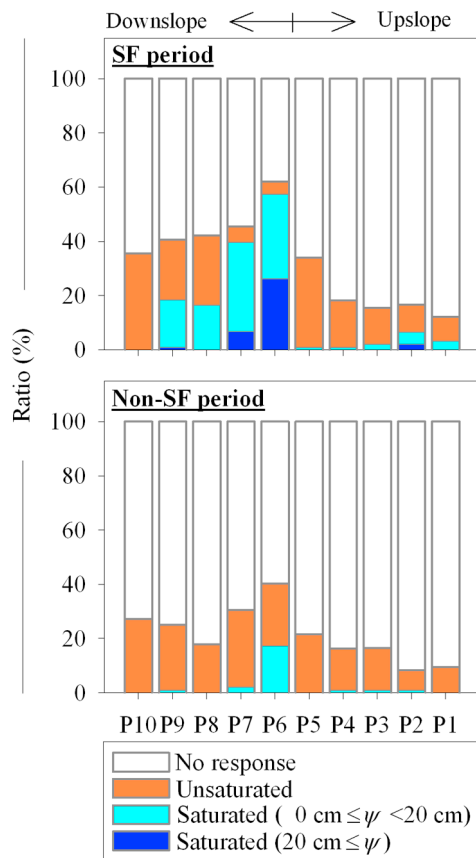


Figure 12. The ratio of rainfall events in which a saturated zone was generated to all rainfall events at each point during the SF and non-SF periods. Saturated zone generation was determined on the basis of the maximum value of pore water pressure (ψ) at the soil-bedrock interface for each rainfall event.

[Liang *et al.*, 2009a], in which stemflow was parameterized as a spring source flux in soil source regions from depths of 5–55 cm near a tree and separately inputted to each 10 cm depth by different ratios. The model exhibited the irregular distribution of vertical soil water content changes and a rapid response in the deeper soil layer, corresponding to results of the dye experiments in this study (Figure 8a). Therefore, we further suggest that inputting stemflow to the source area in soil layers determined by root architecture could simulate soil water dynamics around a tree more accurately.

4.2. Hillslope Hydrological Processes in a Condition Without Stemflow Supply

[38] For the soil water dynamics in the non-SF period, dramatic increases or decreases of θ disappeared at the points close to the downslope side of the tree, and differences in soil water response between upslope and downslope regions became small (Figure 6a). For the detailed soil water dynamics, irregular distributions of $\Delta\theta$ still existed at P6 (Figure 6b). P6 and P7 showed irregular distributions of $\Sigma\Delta\theta^*$ (Figure 9) and relatively high frequency of $\Delta S_{\max}$ greater than accumulated open-area rainfall (Figure 11) in comparison to the other points. Active bypass flow in the downslope region (Figure 10) contributed to a relatively high response ratio of

ψ at the soil-bedrock interface in the downslope region and saturated zone generation at P6 (Figure 12). Thus, asymmetrical soil water responses still existed in the non-SF period but were less than in the SF period.

[39] The asymmetrical infiltration patterns in the non-SF period were highly related to root architecture that diverted infiltrated rainwater into deeper layers. On the basis of sprinkling experiments and fine root samples limited to 0–4.5 mm in diameter, Lange *et al.* [2009] provided convincing evidence that tree roots represent the pore system and carry preferential flow, which improves infiltration. In this study, we found that the relationship of soil porosity and $\Sigma\Delta\theta^*$ was not statistically significant at P6 and P7 but was significant at the other points. In fact, it was not easy to sample soil around the tangled root architecture, particularly at the soil-root interface around thick roots, which are thought to serve as more active and preferential infiltration pathways than fine roots (e.g., P6 and P7).

[40] Preferential infiltration not only affected the spatial distribution of soil water content but also contributed to saturated zone generation at the soil-bedrock interface. Although saturated zone generation at the soil-bedrock interface attributed to stemflow infiltration was reported by Durocher [1990] and in our previous study [Liang *et al.*, 2007], the saturated zone generated in a condition without stemflow had not been reported. Although the amount of saturated zone generated at P6 in the non-SF period was less than that in the SF period, the saturated zone did expand downward and upward in both periods, which increased the response ratios of ψ at the soil-bedrock interface at P7 and P5 (Figure 12). Tromp-van Meerveld and McDonnell [2006b] pointed out that the generation and connectivity of the saturated zone at the soil-bedrock interface were mainly controlled by bedrock microtopography and, furthermore, indicated that vegetation has a larger influence on spatial soil moisture patterns because of spatially variable evapotranspiration [Tromp-van Meerveld and McDonnell, 2006a]. In addition to these previous studies, this study provides evidence that saturated zone generation is highly related to the location of a tree, even in conditions without stemflow supply.

5. Summary and Conclusions

[41] We analyzed the effects of stemflow and root-induced bypass flow on hillslope hydrological processes using 2 years of field observations of soil water dynamics around a tree before (SF period) and after (non-SF period) intercepting stemflow. The results are summarized below.

[42] 1. The soil water dynamics in the SF period showed rapid variations and irregular distributions of soil water content and considerably high soil water storage at points close to the downslope side of the tree, where frequent development of a saturated zone was observed. This result was attributed to the concentration of a large amount of stemflow at the downslope side of the tree. Thus, stemflow caused asymmetrical soil water dynamics between regions downslope and upslope from the tree.

[43] 2. Although the soil water dynamics in the non-SF period showed small differences in soil water response patterns in comparison to the SF period, asymmetrical soil water response patterns were still observed. A relatively higher frequency of bypass flow occurrence, larger soil water storage, and the generation of a saturated zone in the downslope

region from the tree were attributed to root-induced bypass flow.

[44] 3. Two sprinkling experiments using different dye tracers were conducted to separately identify infiltration pathways derived from throughfall and stemflow. Focusing on the point closest to the downslope side of the tree, different bypass pathways from throughfall and stemflow were observed. In particular, the stemflow input bypassed surface soil layers and infiltrated horizontally along thick roots.

[45] 4. Focusing on the downslope region from the SF to non-SF periods, the frequencies of maximum soil water storage greater than the accumulated rainfall decreased from 43% to 10% and saturated zone generation at the soil-bedrock interface decreased from 58% to 16%, but bypass flow occurrence changed little. Therefore, stemflow did contribute to soil water response, soil water storage, and saturated zone generation but not to bypass flow occurrence, which was determined by root architecture.

[46] The development of a saturated zone was observed underneath the tree with or without stemflow supply, indicating that a tree site is a key location for rainfall infiltration. The generation of a saturated zone at the soil-bedrock interface contributing to runoff generation and a deeper bedrock flow maintaining base flow [Montgomery and Dietrich, 2002; Kosugi et al., 2006] hold important information for predicting the location and timing of shallow landslide occurrences [Montgomery et al., 2002; Dhakal and Sidle, 2004]. We suggest that the processes of saturated zones at the soil-bedrock interface should be considered when investigating the effects of a forest on hillslope hydrological processes. In particular, the effects of a tree on subsurface flow and runoff generations that were not clarified in this study could be one direction of further studies.

[47] **Acknowledgments.** The open-area rainfall and environmental data used in this study were provided by the Kamigamo Experimental Station of Kyoto University, Japan. This work was partly supported by grants from the Monbukagakusho Fund for Scientific Research (22248018, 201830, and 19380087) of Japan.

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